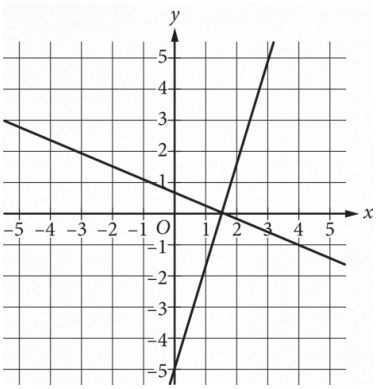
Systems of two linear equations in two variables — Part 2

Algebra · Math · 63 questions

2.1 016627-DC · Medium



Which of the following systems of equations has the same solution as the system of equations graphed above?

A. $y = 0$
$x = \frac{3}{2}$

B. $y = \frac{3}{2}$
$x = 0$

C. $y = 0$
$x = 1$

D. $y = 1$
$x = 0$

2.2 572fa74d-4b79-415a-8241-a2848bbba074 · Medium

$$y = 6x + 16$$

$$-7x - y = 36$$

What is the solution x, y to the given system of equations?

A. $-4, -8$

B. $-\frac{20}{13}, -\frac{80}{13}$

C. $4, 40$

D. $20, 136$

2.3 09372-DC · Medium

A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

A. 30

B. 20

C. 19

D. 18

2.4 181b5ada-d520-4724-9962-4d99a5d79e8f · Medium

$$3y = 4x + 17$$

$$-3y = 9x - 23$$

The solution to the given system of equations is x, y. What is the value of $39x$?

A. -18

B. -6

C. 6

D. 18

2.5 cb6510bf-691c-4361-8c92-372e7946de91 · Medium

$$y = -\frac{1}{5}x$$

$$y = \frac{1}{7}x$$

The solution to the given system of equations is x, y. What is the value of x?

A. -5

B. 0

C. 2

D. 7

2.6 e8d1c0cc-b06d-4425-92fb-3beba3af4800 · Medium

$$24x + y = 48$$

$$6x + y = 72$$

The solution to the given system of equations is (x, y). What is the value of y?

Student-produced response: _____

2.7 7f5e8f0d-75c3-4c2d-ac2a-90f3c586f44c · Medium

$$y - 9x = 13$$

$$5x = 2y$$

What is the solution x, y to the given system of equations?

A. $\frac{5}{2}, 1$

B. $1, \frac{2}{5}$

C. $-2, -5$

D. $-5, -2$

2.8 a9ea688b-0693-4bef-8150-b731185309a2 · Medium

Which of the following systems of linear equations has no solution?

A. $x = 3$ $y = 5$

B. $y = 6x + 6$ $y = 5x + 6$

C. $y = 16x + 3$ $y = 16x + 19$

D. $y = 5$ $y = 5x + 5$

2.9 025580-DC · Medium

$$5x + 3y = 38$$
$$x + 3y = 10$$

In the solution (x, y) to the system of equations above, what is the value of x ?

Student-produced response: _____

2.10 034cf40e-98a7-4817-a3d0-a76cae0faf07 · Medium

$$y = 3x + 9$$

$$3y = 8x - 6$$

The solution to the given system of equations is x, y. What is the value of $x - y$?

A. -123

B. -33

C. 3

D. 57

2.11 f02c7424-034b-4774-b41c-847249027a2e · Medium

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was 1,440 dollars, how many people in the group were children?

A. 3

B. 9

C. 12

D. 18

2.12 04476-DC · Medium

The score on a trivia game is obtained by subtracting the number of incorrect answers from twice the number of correct answers. If a player answered 40 questions and obtained a score of 50, how many questions did the player answer correctly?

Student-produced response: _____

2.13 10551-DC · Medium

$\frac{1}{2}y = 4$
$x - \frac{1}{2}y = 2$

The system of equations above has solution (x, y). What is the value of x ?

A. 3

B. $\frac{7}{2}$

C. 4

D. 6

2.14 87828f3a-da5a-4eeb-9cec-04f0ff9e1f97 · Medium

$$-15x + 25y = 65$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which of the following could be the second equation in the system?

A. $12x + 20y = 52$

B. $12x + 20y = -52$

C. $-12x + 20y = 52$

D. $-12x + 20y = -52$

2.15 a87ec507-656c-4bf5-ad90-09107b7dcf8b · Medium

$$8x + y = 5$$

$$y = 9x + 1$$

The solution to the given system of equations is x, y. What is the value of x?

A. -6

B. $\frac{4}{17}$

C. $\frac{6}{17}$

D. 4

2.16 33e5c79f-6eb4-44bc-917c-7090df8d4158 · Medium

Which of the following systems of linear equations has no solution?

A. $y = 6x + 3$ $y = 6x + 9$

B. $y = 10$ $y = 10x + 10$

C. $y = 14x + 14$ $y = 10x + 14$

D. $x = 3$ $y = 10$

2.17 42dab414-d361-4e77-a550-5863631c999e · Medium

$$y - 6 = \frac{4}{5}x + 5$$

$$y - v = \frac{4}{5}x + 10$$

In the given system of equations, v is a constant. The system has infinitely many solutions. What is the value of v?

A. 0

B. 2

C. 4

D. 12

2.18 aa6865b9-fa65-43d3-bbf6-6e17250dc546 · Medium

$$x + 3y = 29$$

$$3y = 11$$

The solution to the given system of equations is x, y. What is the value of x?

Student-produced response: _____

2.19 765f83e6-6d57-4694-82d5-9b2180d80565 · Medium

$$y = 4x$$

$$5x - 3y = -14$$

The solution to the given system of equations is x, y. What is the value of $x + y$?

A. 4

B. 6

C. 8

D. 10

2.20 11158-DC · Medium

$$4x + 5y = 100$$
$$5x + 4y = 62$$

If the system of equations above has solution (x, y), what is the value of $x + y$?

A. 0

B. 9

C. 18

D. 38

2.21 b992e5d3-e84d-4c00-86c0-75f34b0c6f14 · Medium

$$y = -\frac{1}{9}x$$

$$y = \frac{1}{2}x$$

The solution to the given system of equations is x, y. What is the value of x?

A. -9

B. -7

C. 0

D. 2

2.22 ab4fb25b-f569-45af-8536-c3264af6ff16 · Hard

$$ax + by = 72$$

$$6x + 2by = 56$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy-plane intersect at the point $4, y$. What is the value of a?

A. 3

B. 4

C. 6

D. 14

2.23 e124013e-471c-4f28-a3b4-212a4fd25c4e · Hard

Which system of linear equations has no solution?

A. $-2x + 3y = -9$ $2x - 3y = 9$

B. $2x - 3y = 9$ $3x + 4y = 10$

C. $2x - 3y = 9$ $-6x + 9y = -27$

D. $-2x + 3y = 9$ $4x - 6y = 18$

2.24 bc24a37b-b601-44de-a222-60974e9db0fb · Hard

Store A sells raspberries for $\$ 5.50$ per pint and blackberries for $\$ 3.00$ per pint. Store B sells raspberries for $\$ 6.50$ per pint and blackberries for $\$ 8.00$ per pint. A certain purchase of raspberries and blackberries would cost $\$ 37.00$ at Store A or $\$ 66.00$ at Store B. How many pints of blackberries are in this purchase?

A. 4

B. 5

C. 8

D. 12

2.25 15e8ec7c-ca9d-47c1-ab6b-a40b86f74651 · Hard

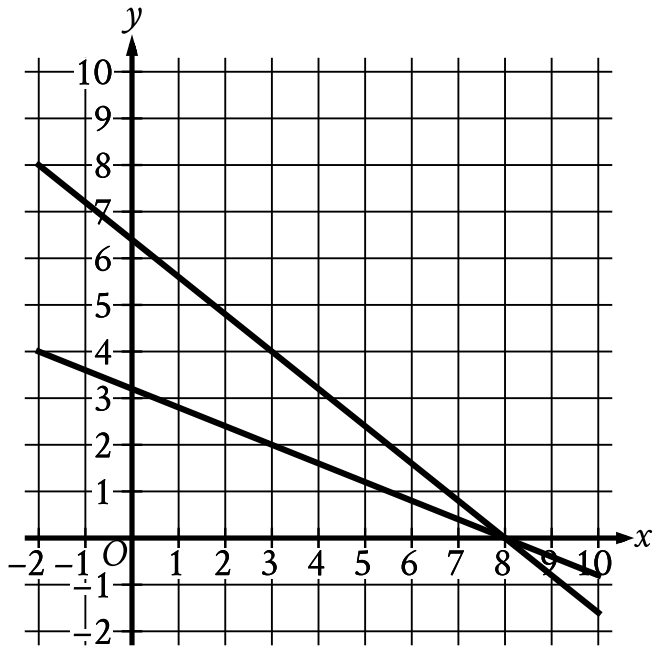
$$x - 2 - 4y + 7 = 117$$

$$x - 2 + 4y + 7 = 442$$

The solution to the given system of equations is x, y. What is the value of $6x - 2$?

Student-produced response: _____

2.26 c240c048-e272-462b-b645-1751de050cdd · Hard



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (3 comma 2)
    - (8 comma 0)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (3 comma 4)
    - (8 comma 0)

What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$ $-10x - 4y = -64$

B. $8x - 4y = 32$ $-10x + 4y = -64$

C. $4x - 10y = 32$ $-8x + 10y = -64$

D. $4x + 10y = 32$ $-8x - 10y = -64$

2.27 031d354d-96ee-46e5-b453-c0be0ad459d5 · Hard

$$3x = 36y - 45$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

A. $x = 4y$

B. $\frac{1}{3}x = 4y$

C. $x = 12y - 15$

D. $\frac{1}{3}x = 12y - 15$

2.28 e07c6477-a704-4fb6-8142-4f45390607dc · Hard

In August, a car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. In August and September, the car dealer completed 363 sales. How many sales did the car dealer complete in September?

Student-produced response: _____

2.29 673c8b0c-840e-41ac-bc4e-82b2f8b2a747 · Hard

$$y = 6x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

A. $-6x + y = 18$

B. $-6x + y = 22$

C. $-12x + y = 36$

D. $-12x + y = 18$

2.30 060e402c-37f7-47bb-8481-4930b50fde96 · Hard

$$y = 7x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

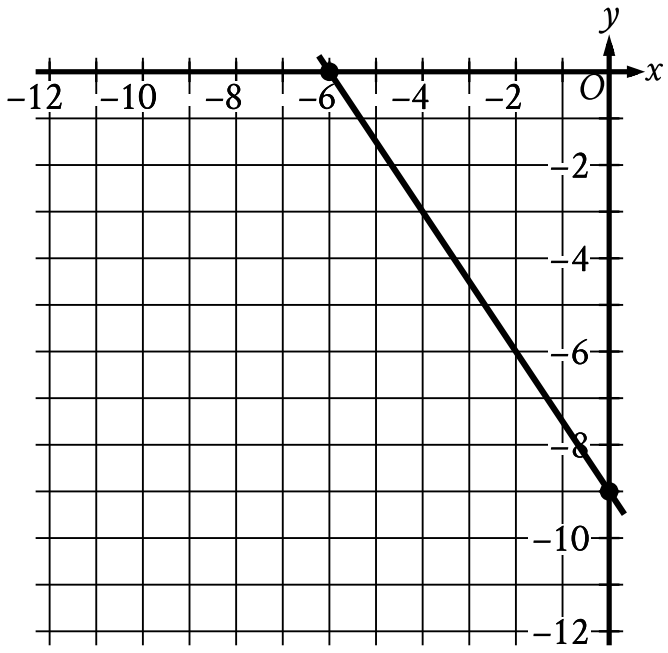
A. $-21x + y = 54$

B. $-21x + y = 18$

C. $-7x + y = 21$

D. $-7x + y = 18$

2.31 bf3d7c38-346b-4b31-8065-9d69383f8a13 · Hard



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (negative 6 comma 0)
  - (0 comma negative 9)

The graph of line h is shown in the xy-plane. Line k (not shown) is defined by $sx + 48y = t$, where s and t are constants. If line k is graphed in this xy-plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is NOT a possible value of t?

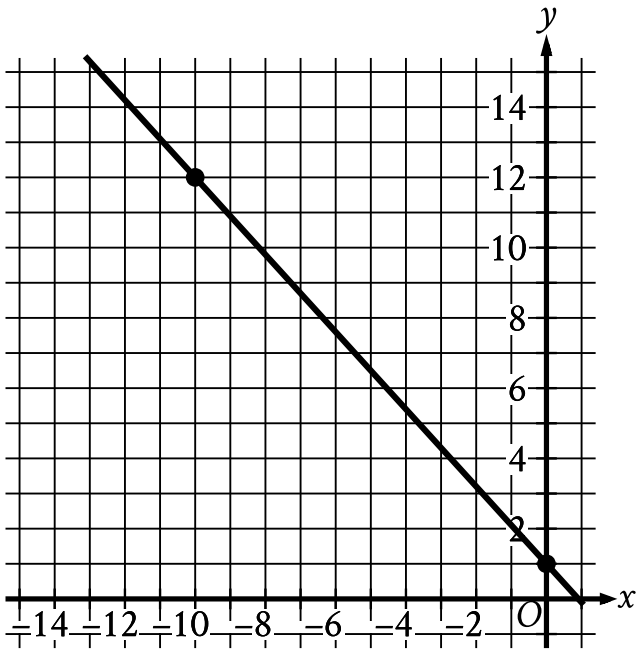
A. -432

B. -9

C. 72

D. 288

2.32 e9d39168-6488-45c7-828e-71039f10ceca · Hard

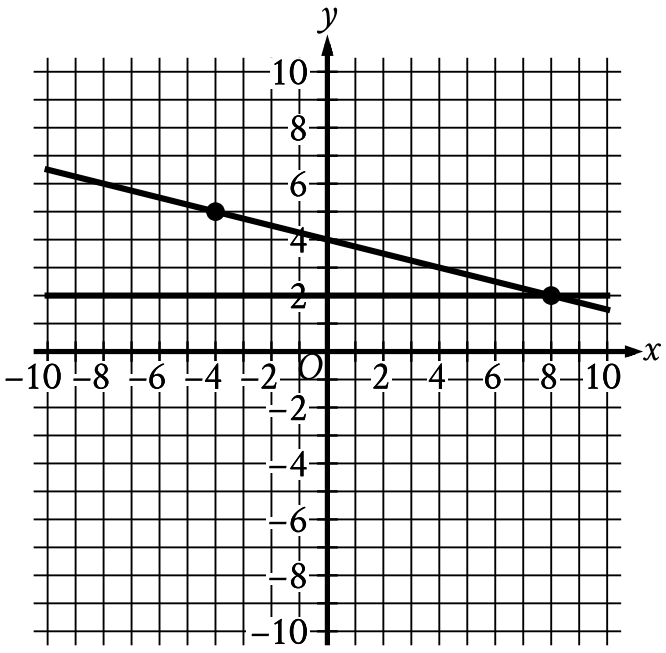


- The line slants sharply down from left to right.
- The line passes through the following points:
  - (negative 10 comma 12)
  - (0 comma 1)

The graph of line g is shown in the xy-plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this xy-plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

Student-produced response: _____

2.33 f6ec6740-9074-4a1f-a62d-cac86fd126d8 · Hard



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (negative 4 comma 5)
    - (0 comma 4)
    - (8 comma 2)
- For the second line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (0 comma 2)
    - (8 comma 2)

If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions x, y will the resulting system of three equations have?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

2.34 3bd8d2aa-c0a4-4f79-8d4e-5eb3b5c814f0 · Hard

$$6 + 7r = pw$$

$$7r - 5w = 5w + 11$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Student-produced response: _____

2.35 64207213-df09-436f-aeeb-a588aafa2b18 · Hard

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r?

Student-produced response: _____

2.36 029346-DC · Hard

$$7x - 5y = 4$$
$$4x - 8y = 9$$

If (x,y) is the solution to the system of equations above, what is the value of $3x + 3y$?

A. -13

B. -5

C. 5

D. 13

2.37 029548-DC · Hard

During a month, Morgan ran r miles at 5 miles per hour and biked b miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

A. 80

B. 100

C. 120

D. 160

2.38 82f7e038-e8ce-4d33-b043-0d0d68435a80 · Hard

$$5y = 10x + 11$$

$$-5y = 5x - 21$$

The solution to the given system of equations is x, y. What is the value of $30x$?

Student-produced response: _____

2.39 070616-DC · Hard

$$y = \frac{1}{2}x + 8$$

$$y = cx + 10$$

In the system of equations above, c is a constant. If the system has no solution, what is the value of c ?

Student-produced response: _____

2.40 034116-DC · Hard

One of the two equations in a linear system is $2x+6y=10$. The system has no solution. Which of the following could be the other equation in the system?

A. $x+3y=5$

B. $x+3y=-20$

C. $6x-2y=0$

D. $6x+2y=10$

2.41 7f109a34-c477-4f2b-9704-3ccd5306ad84 · Hard

$$-12x+14y=36$$

$$-6x+7y=-18$$

How many solutions does the given system of equations have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.42 5c0525ca-3f74-406b-82bd-232050622e14 · Hard

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A. $r, -\frac{6r}{7} + \frac{5}{7}$

B. $r, \frac{7r}{6} + \frac{5}{6}$

C. $\frac{r}{4} + 5, -\frac{r}{4} + 20$

D. $-\frac{6r}{7} + \frac{5}{7}, r$

2.43 96dd952c-8f39-4328-a087-4989624fb279 · Hard

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is x, y. What is the value of $x - y$?

Student-produced response: _____

2.44 b07eee5e-2b48-42ca-9bdf-9cdd01089b62 · Hard

A sample of a certain alloy has a total mass of 50.0 grams and is 50.0% silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. What was the mass, in grams, of the silicon in the second piece?

A. 9.0

B. 16.0

C. 20.0

D. 30.0

2.45 9961525f-7f16-4e8c-9aef-754d7403a0cf · Hard

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h?

A. -9

B. 0

C. 9

D. 18

2.46 8b3e73a1-172a-4ae8-af4d-a8d63b197417 · Hard

$$2x + 9y = 7$$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

I. $3x + 13.5y = 10.5$
II. $3x - 13.5y = 10.5$

A. I only

B. II only

C. I and II

D. Neither I nor II

2.47 885fbade-61fe-4b3d-b976-1a5abfa9ae0c · Hard

$$38(x - n) = 38y + 38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

A. $4x - 4y = 8n$

B. $4x + 4y = 4n$

C. $4x + 4y = 8n$

D. $4x - 4y = 4n$

2.48 99370f6d-8788-4824-a8aa-20c268f5a86c · Hard

$$2x + 3y = 7$$
$$10x + 15y = 35$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A. $\frac{r}{5} + 7, \ -\frac{r}{5} + 35$

B. $-\frac{3r}{2} + \frac{7}{2}, \quad r$

C. $r, \quad \frac{2r}{3} + \frac{7}{3}$

D. $r, \ -\frac{3r}{2} + \frac{7}{2}$

2.49 f55f83d0-fd97-488f-9f5c-014bc1e34d9c · Hard

$$\frac{2}{5}x + \frac{7}{5}y = \frac{2}{7}$$

$$gx + ky = \frac{5}{2}$$

In the given system of equations, g and k are constants. The system has infinitely many solutions. What is the value of $\frac{g}{k}$?

Student-produced response: _____

2.50 35175215-d90c-49a0-927c-7f816dae6a5d · Hard

$$x + y = 20$$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

A. $3x + 3y = 60$

B. $-x - y = 40$

C. $3x + 3y = 14$

D. $x - y = 20$

2.51 872b58e5-a9ea-488a-917d-0bdaaf2a4d73 · Hard

$$\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$$

$$\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Student-produced response: _____

2.52 37c0d421-25f7-4ad6-99d7-8c81927bd65c · Hard

A piece of wire with a length of 32 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 4 more than 3 times the value of y. What is the value of x?

Student-produced response: _____

2.53 f1fcd45c-00d6-4477-bbcd-75ebc3cfc227 · Hard

$$48x - 64y = 48y + 24$$

$$ry = \frac{1}{8} - 12x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r?

Student-produced response: _____

2.54 070615-DC · Hard

$$-x + y = -3.5$$
$$x + 3y = 9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

Student-produced response: _____

2.55 05268-DC · Hard

A movie theater sells two types of tickets, adult tickets for $12 and child tickets for $8. If the theater sold 30 tickets for a total of $300, how much, in dollars, was spent on adult tickets? (Disregard the $ sign when gridding your answer.)

Student-produced response: _____

2.56 022887-DC · Hard

$y = 2x + 1$
$y = ax - 8$

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

A. $-\frac{1}{2}$

B. 0

C. 1

D. 2

2.57 05435-DC · Hard

In the system of equations below, a and c are constants.

$\frac{1}{2}x + \frac{1}{3}y = \frac{1}{6}$
$ax + y = c$

If the system of equations has an infinite number of solutions (x, y), what is the value of a ?

A. $-\frac{1}{2}$

B. 0

C. $\frac{1}{2}$

D. $\frac{3}{2}$

2.58 d384e4e9-870d-4026-ba15-b7eb23e0c5fc · Hard

$$28x + 47y = 12$$

$$-28x + 47y = 12$$

The solution to the given system of equations is x, y. What is the value of $8x + 7y$?

Student-produced response: _____

2.59 386cf16e-ddc8-4613-85b9-de02ba844834 · Hard

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Student-produced response: _____

2.60 425d30d9-246b-4058-b45c-2e4168f92a0f · Hard

$$5x + 14y = 45$$

$$10x + 7y = 27$$

The solution to the given system of equations is x, y. What is the value of xy?

Student-produced response: _____

2.61 c4e7a98c-fecd-45d1-ad5b-69c978af6bf7 · Hard

$$8x + 7y = 9$$
$$24x + 21y = 27$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A. $r, -\frac{8r}{7} + \frac{9}{7}$

B. $-\frac{8r}{7} + \frac{9}{7}, r$

C. $-\frac{8r}{7} + 9, \frac{8r}{7} + 27$

D. $\frac{r}{3} + 9, -\frac{r}{3} + 27$

2.62 3265ed42-e3f1-44ac-9cb6-a8b55c776c51 · Hard

$$-x - wy = -337$$

$$2x - wy = 47$$

In the given system of equations, w is a constant. In the xy-plane, the graphs of these equations intersect at the point $q, 19$, where q is a constant. What is the value of w?

Student-produced response: _____

2.63 5c09d547-9da9-4b9a-a436-ddc8c0c70764 · Hard

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$$

$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Student-produced response: _____
